

PARV RUPESH GHEEWALA

[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

Gujarat, India · +91 7069105031 · gheewalaparv@gmail.com

Backend Engineer — Distributed Systems — AI Infrastructure — Open Source

Summary

Backend engineer passionate about building reliable, scalable, and distributed systems. Experience building multilingual AI infrastructure and high-concurrency backend services across four software engineering internships.

Education

VIT Vellore, B.Tech in Computer Science

Sept 2022 – July 2026

CGPA: 8.51/10

Patents

Indian Patent Application – Indian Patent Office

2026

Election Grade Mobile E-Voting System (A Scalable, Verifiable, and Sovereign Voting Architecture)

Co-Inventor with Dr. Goutam Majumder (Assistant Professor, VIT)

IPR ID: 202641072294 · Status: Patent Published

Open Source Contributions

Bharat2Braille – Bidirectional Translation for Indian Languages – [GitHub](#)

Aug 2025 – Dec 2025

- Extended one-way Braille encoder by implementing bidirectional translation module enabling Braille ↔ text conversion for **9 Indian languages** (Hindi, Marathi, Gujarati, Punjabi, Sanskrit, Tamil, Telugu, Kannada, Malayalam), empowering **2.6M+** Braille readers.
- Developed reverse mapping with context-aware disambiguation for Bharati Braille contractions, achieving **95%** reconstruction accuracy validated by Tenpoints Braille.

Experience

Full-Stack Development Intern – AI4Bharat, Remote

Jan 2026 – Jun 2026

- Implemented Thai and Indonesian multilingual support across chat, TTS, ASR.
- Migrated backend from WSGI to hybrid ASGI–WSGI architecture.
- Built scalable multilingual production systems for low-resource language AI.

Software Development Intern – BloomSocialAI, Remote

Apr 2026 – May 2026

- Built core product features and a hashtag recommendation system for the BloomSocialAI UI, supporting **100+** paid users.
- Load-tested backend services, achieving **5K+** concurrency with sub-**20ms** average latency.
- Leveraged caching and efficient API design to maintain consistent low-latency performance under scale.

Full-Stack Developer Intern – Code Space Techlabs, Remote

Aug 2025 – Dec 2025

- Optimized Node.js/FastAPI services with Redis caching and PostgreSQL indexing, reducing API latency by **60%** for **40K+** daily requests.
- Built responsive React dashboards with lazy loading, improving initial load time by **45%** and Core Web Vitals scores.

Frontend Developer Intern – FameUX, Surat

Jun 2024 – Jul 2024

- Reduced page load times using lazy loading and bundle optimization, cutting bundle size by **38%**.

Projects

InvestRight – Mutual Fund Analytics – [GitHub](#)

Aug 2025 – Sept 2025

React, Vite, FastAPI, PostgreSQL, Redis, scikit-learn, Docker

- Built FinTech platform serving **14,000+** mutual funds with **21M+** NAV records and **14K+** daily updates via distributed scheduler.
- Implemented ML engine (Random Forest, Gradient Boosting, K-Means) with periodic retraining for personalized recommendations.
- Optimized FastAPI with PostgreSQL indexing and Redis caching, reducing latency by **40%** (850ms → 510ms).

Cronify – Distributed Task Scheduler – [GitHub](#)

Feb 2024 – May 2024

FastAPI, Redis Streams, PostgreSQL, Docker, Next.js, Prometheus, Grafana

- Built distributed scheduler with **99.8%** delivery success and sub-**500ms** latency for **1,000+** concurrent jobs.
- Designed fault-tolerant pipeline using Redis Streams with exponential backoff, HMAC-SHA256 auth, and dead letter queues.

AI Customer Support Chatbot – [GitHub](#) · [Live](#)

Oct 2025

TinyLlama-1.1B, Sentence-BERT, FastAPI, Pinecone, LangChain, React, Docker

- Built RAG chatbot over **500+** documents, reducing query resolution time from **30s** to **3s**.
- Added semantic caching with Pinecone, cutting LLM API costs by **40%** while maintaining **99.5%** uptime.

Skills

Languages: Python, JavaScript, SQL, Java

Frontend: React, Next.js, Tailwind CSS, shadcn/ui

Backend: FastAPI, Node.js, REST APIs, JWT/OAuth, Microservices

Databases: PostgreSQL, MongoDB, Redis, SQLAlchemy

AI/ML: LangChain, RAG, Vector DBs (Pinecone, pgvector), Sentence-BERT, PyTorch, scikit-learn

DevOps: Git, Docker, CI/CD, AWS, Railway, Vercel, Prometheus, Grafana

Certifications

Oracle: OCI 2025 Generative AI Professional, Multicloud Architect Professional [1] [2]

ActiveLoop: LangChain & Vector DBs [3]

DeepLearning.ai: Generative AI with LLM [4], NLP Specialization [5], MLOps Specialization (ongoing)

Achievements

- Ranked **Top 8%** (400/5000) in VIT Top Coder competitive programming contest.
- Solved **300+** DSA problems on Leetcode and GeeksforGeeks.